Student Name: _____ Date: _____
Class: Grade 11 / _____ | **Aim High 5** Score: _____ / 20

UNIT 3 — READING — THE ORIGIN OF INSPIRATION & ARCHIMEDES

Focus: The Origin of Inspiration (Archimedes & Eureka)

LEVEL: STANDARD / INDEPENDENT PRACTICE

Part 1: Key Vocabulary & Terminology

Match the correct vocabulary term or definition:

1. An important discovery or sudden advance that leads to success — **Term:** _____
2. To move water or an object out of its normal position — **Term:** _____
3. The degree of compactness and mass of a substance per volume — **Term:** _____
4. To formally authorize or request someone to make or design something — **Term:** _____
5. Feeling or showing a belief that someone is guilty of dishonest behavior — **Term:** _____

Part 2: Comprehension & Concept Application

Choose the correct option according to the lesson:

1. Why did the King of Syracuse commission Archimedes to examine his new royal crown?
   A) He suspected the goldsmith had secretly replaced pure gold with cheaper silver
   B) He wanted to know if the crown was strong enough for battlefield combat **C)** He wished to sell it to an emperor in Rome
   D) He wanted to duplicate it into ten identical crowns
2. Where did the flash of inspiration occur to Archimedes after weeks of frustration?
   A) While lowering himself into a public bath **B)** While walking through an olive grove at sunset
   C) While reading scrolls in the Alexandria library **D)** While listening to a theatrical tragedy
3. How did observing the bath water rising and spilling over solve the king's puzzle?
   A) It proved that an object submerging displaces a volume of water equal to its own volume
   B) It showed that pure gold dissolved instantly in warm water **C)** It caused the crown to float on top of the surface
   D) It cleaned the silver oxidation away from the metal
4. What did Archimedes exclaim as he leaped from the bath and ran through the streets?
   A) Eureka! (I have found it!) **B)** Victory is ours! **C)** Glory to Syracuse! **D)** Truth prevails!

Part 3: Practical Sentence Completion

Complete each sentence with a grammatically accurate term from the unit:

1. Archimedes achieved an extraordinary scientific _____ (breakthrough) in Syracuse.
2. The king was deeply _____ (suspicious) that he had been cheated by the dishonest craftsman.
3. Because silver has a lower _____ (density) than gold, it displaced a larger volume of water.